Utkarsh Sharma

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Education

Indian Institute of Science (IISc) — **Bangalore, India**
M.Tech in Computer Science and Engineering | ***CGPA: 8.7/10.0*** — *2024 – 2026*

Key Coursework: Deep Learning for CV, Learning for 3D Vision and Inverse Graphics, Graphics & Visualization, Computational Methods of Optimization, Machine Learning for Signal Processing, Design and Analysis of Algorithms

Chandigarh University — **Mohali, India**
B.E. in Computer Science | ***CGPA: 8.24/10.0*** — *2020 – 2024*

Technical Skills

Languages: Python, C++, CUDA, SQL
Computer Vision: Physically Based Rendering (PBR), Neural Path Guiding, Neural Rendering, NeRFs, 3DGS
Deep Learning: Vision Transformers, Autoencoders, Normalizing Flows, Diffusion Models, Self-Supervised Learning
Frameworks & Tools: PyTorch, NumPy, Mitsuba 3, PBRT, Tiny-CUDA-NN, Dr.Jit, Pandas, Git

Research & Projects

Neural Path Guiding for Path Tracing | *Mitsuba 3, PyTorch, Tiny-CUDA-NN* — **June – Dec 2025**

- Implemented path guiding techniques like **Neural Parametric Mixtures (NPM)**, **Neural Importance sampling (NIS)**, **Distribution Factorization (DF)** for guiding the ray direction in Path Tracing reducing the variance by $2-4\times$ compared to BRDF sampling.
- Implemented **Neural Radiance Caching** for real-time global illumination approximation, using hierarchical spatial hash encoding and online gradient-based optimization during the render loop.

CUDA Path Tracer with Monte Carlo Radiosity | *C++, CUDA, Python* — **Aug – Dec 2025**

- Developed a renderer containing both unbiased **Path Tracing** and **Radiosity** for physically accurate diffuse inter-reflections and caustic light transport.
- Implemented custom **Bounding Volume Hierarchy (BVH)** for faster ray intersection speed.
- Integrated multiple importance sampling (MIS) for combining BSDF and light sampling strategies, reducing fireflies and noise in difficult lighting conditions.

Masked Autoencoder (MAE) using Vision Transformer (ViT) | *PyTorch, Self-Supervised Learning* — **Oct – Dec 2025**

- Implemented a Masked Autoencoder with a transformer encoder-decoder architecture.
- Trained the model to reconstruct missing image patches from latent representations, enabling self-supervised pre-training
- Visualized attention maps to interpret how the model aggregates context from unmasked patches to predict occlusion.

Deep Learning-Based Denoising for Monte Carlo Rendering | *PyTorch, PBRT* — **Feb – Apr 2025**

- Developed and benchmarked **diffusion-based and U-Net denoisers** for low-sample Monte Carlo path tracing outputs
- Incorporated **G-Buffer auxiliary features** (albedo, normals, depth, motion vectors) as multi-channel inputs to guide network predictions and preserve geometric edges and material boundaries.
- Achieved superior perceptual quality (SSIM $>$ 0.95, PSNR $>$ 35 dB) on PBRT-generated test scenes

Neural Radiance Fields & 3D Gaussian Splatting | *PyTorch* — **Aug – Dec 2025**

- Built a **NeRF pipeline from scratch** using PyTorch, implementing volumetric rendering equations, sinusoidal positional encoding, and hierarchical stratified sampling to reconstruct complex 3D scenes.
- Implemented **3D Gaussian Splatting**, optimizing 3D covariance matrices and spherical harmonics (SH) coefficients.
- Achieved high-fidelity reconstruction on the **NeRF Synthetic dataset** (PSNR $>$ 25 dB), validating the results.

Teaching Experience

Teaching Assistant – Algorithms & Programming | *Indian Institute of Science* — **Aug 2025 – Dec 2025**
Teaching Assistant – Computational Topology | *Indian Institute of Science* — **Jan 2026 – Present**

Academic Achievements

Rank 8 out of 123,967 candidates in GATE Computer Science (CS) 2024
Rank 17 out of 39,210 candidates in GATE Data Science & AI (DA) 2024
Winner, CodeSmash 2.0 Competitive Programming Contest
Finalist, CodeRush 3.0 Competitive Programming Contest